

**PROGRAMMING ASSIGNMENT 1****Due date:** 21.03.2025 23:00

In this assignment, you will run a set of benchmark programs in the [National Center for High Performance Computing \(UHeM\)](#) at Istanbul Technical University (ITU).

The steps to be completed in the assignment as follows:

- 1) [Register](#) for access to the UHEM as an undergraduate student. You need to fill in the [application form](#) and await for the confirmation. Get familiar with the system through [UHEM wiki pages](#) to understand the system and how to use it.
- 2) After getting user name and password, connect to the UHEM system by using the [instructions](#).
- 3) Download Polybench C benchmark that can be found in the following link: <https://www.cs.colostate.edu/~pouchet/software/polybench/download/polybench-c-3.2.tar.gz> (You can use [wget](#) command).
- 4) Extract the downloaded file (You can use [tar](#) command).
- 5) Select five programs from the available applications. Compile each benchmark program (e.g., [datamining/correlation](#)) by using the appropriate commands given in README file.
- 6) Prepare shell scripts to execute each program with default parameters on the remote resources 10 times, collect the execution time of each execution (with [time](#) command) and take average (of real times). Report the average execution time for each benchmark in a text file. For the execution of each program, collecting execution times, taking average, and writing into output file, you are required to write two shell scripts:
  - **Compute script** (compute.sh) is to be launched via the slurm script file (given as part of assignment). It will be responsible for executing each benchmark program 10 times on the compute node, measuring the execution time for each run, and storing these times in the output\_shell.txt.
  - **Process script** (process.sh) is to be run in the login node. After the SLURM job completes, you will execute this script. It will read the execution times from the file (output\_shell.txt) generated on the compute node, calculate the average, and write the result to output\_times.txt. This script takes the path to the execution times file as an argument.
- 7) By using the slurm script provided as part of the assignment (slurm\_sample.sh), execute the job in the compute node (Use [sbatch](#) command).
- 8) After the SLURM job completes, check the output\_shell.txt file content to understand the execution is completed successfully.
- 9) Run the Process script in the login node (Use normal script execution like ./process.sh). In the output file (output\_times.txt), you will need to see the 5 program names and the corresponding average execution time as follows:

```
correlation 3s
3mm 1s
...
```

**10)** Write a report that explains all the commands in the previous steps. With your report and given script files, someone (already connected to the system-so that you can skip the UHEM registration and connection steps-) should reproduce all the steps.

**Notes:**

- UHEM Wiki page can help you to work on the system:  
[https://wiki.uhem.itu.edu.tr/w/index.php/UHeM\\_Wiki](https://wiki.uhem.itu.edu.tr/w/index.php/UHeM_Wiki)
- Remember it will take time to get familiar with the UHEM system, so start early.
- You need to work individually, no group work is allowed.
- You must submit the two script files and your report.